SABONG

Gamefowl Management and Record System

Software Requirement

Specification

Version 1.00

Revision History

Date	Version	Description	Author
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Software Requirements Specification

1. Introduction

This section introduces the Gamefowl Management Recquirements Specifications.

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to comprehensively define the functional and non-functional requirements for the Gamefowl Management and Record System. It serves as the primary blueprint for developers, project managers, and stakeholders to ensure the final product meets the operational needs of gamefowl breeders and farm owners.

1.1 Scope

The Gamefowl Management and Record System is a robust cloud-based software application designed to automate and organize farm operations. The system encompasses user authentication, detailed individual bird registration and profiling, automated age and status tracking, extensive breeding and lineage trees, health and vaccination scheduling, farm supply inventory management, and match/tournament bracket organization. The system is designed to replace traditional paper-based recording, offering real-time data retrieval across mobile and desktop devices.

1.2 Definitions, Acronyms, and Abbreviations

Gameflow	Chikens Specifically Bred for exhibition or competition
Broodcock / sire	The male parent used specifically for breeding
Broodhen	The female parent used specifically for breeding
Offspring	The direct descendacts or chicks of a specific pairing

1.4 References

The references are:

- Gamefowl Management and Record System - Project Overview Document
- Gamefowl Management UI Mockups and Flowcharts
- IEEE Std 830-1998, IEEE Recommended Practice for SRS

1.5 Overview

The remainder of this document contains a detailed description of the Gamefowl Management and Record System. Section 2 provides a high-level overview of the user environment and operational context. Section 3 exhaustively details the specific functional requirements (categorized by major system features), followed by non-functional criteria including usability, reliability, performance, security, design constraints, and external interfaces.

2. Overall Description

Gamefowl breeders and farm operators traditionally rely on fragmented systems—such as notebooks, whiteboards, or disjointed spreadsheets—to track the lifecycles of their birds. This leads to data loss, missed vaccination schedules, inaccurate lineage tracing, and inventory mismanagement.

The Gamefowl Management and Record System provides a centralized, secure, and accessible digital platform. It caters to varying user types, from individual hobbyists needing simple logging, to large-scale farm owners requiring multi-pen management and intricate pedigree trees. The system is highly interactive, requiring a stable internet connection, and is designed with a mobile-first approach to accommodate users inputting data directly from the field.

3. Specific Requirements

The specific requirements are –

3.1 Functionality

The system provides comprehensive modules for managing users, tracking bird profiles, tracing lineage, maintaining health records, and scheduling matches.

3.1.1 User Authentication and Account Management

3.1.1.1 User Registration - The system shall allow new users to create an account by providing an email address, password, and farm name.

3.1.1.2 User Login/Logout - The system shall authenticate users via email and password, granting access to their specific farm database, and allow secure session termination.

3.1.1.3 Password Recovery - The system shall provide a "Forgot Password" workflow via email verification link.

3.1.2 Chicken Management

3.1.2.1 Add New Chicken Registration - Users can add a new bird by inputting breed, bloodline, sex, color, hatch date, and uploading an image. The system will auto-generate a unique ID.

3.1.2.2 Update Profile and Status - Users can edit existing bird profiles to update their weight, physical traits, or status (e.g., Active, Sold, Deceased, Brood).

3.1.2.3 Pen Assignment - Users can assign, transfer, or unassign birds from specific numbered or named pens on the farm.

3.1.2.4 Dynamic Age Tracking - The system shall automatically compute and display the exact age in months and years based on the recorded hatch date.

3.1.3 Breeding and Lineage

3.1.3.1 Record Breeding Pair - Users can create a breeding record by selecting a Sire and a Dam from their active inventory, logging the mating date.

3.1.3.2 Track Offspring - Users can log the number of eggs, hatch rate, and link newly registered chicks directly to a specific breeding record.

3.1.3.3 View Pedigree Tree - The system shall generate a visual, multi-generational lineage tree showing a bird's ancestry based on linked Sires and Dams.

3.1.4 Health and Supply Tracking

3.1.4.1 Log Treatment/Vaccination - Users can record the administration of vaccines, vitamins, or medications to individual birds or entire pens.

3.1.4.2 Set Care Reminders - Users can schedule future care events (e.g., "Deworm in 3 weeks") which will trigger dashboard alerts.

3.1.4.3 Manage Supply Inventory - Users can add stock levels for feeds and medicines, deduct quantities used, and view current inventory balances.

3.1.5 Match and Event Management

3.1.5.1 Schedule Match - Users can add upcoming derby or sparring dates, assigning specific birds to the event.

3.1.5.2 Record Outcomes - Users can log the result of a match (Win, Loss, Draw) which will automatically update the bird's overall win-rate statistics.

3.1.5.3 Generate Bracket - For organized events, the system shall allow users to build and update tournament brackets as matches progress.

3.1.6 Roster List View

3.1.6.1 Roster Overview Display - The system shall display an overview showing each rooster’s photo, assigned pen tag, bloodline, age, and win-loss record.

3.1.7 Profile Setup

3.1.7.1 Custom Configurations - The system shall support custom photo uploads, rooster naming, and pedigree notes.

3.1.8 Record Summary

3.1.8.1 Detailed Record Breakdown - The system shall provide a detailed breakdown of wins, losses, draws, win rate, birthdate, age, breeder, and pedigree.

3.1.9 Report

3.1.9.1 Comprehensive Reporting - The system shall generate chicken, health, breeding, supply, and event reports.

3.1.10 Artificial Intelligence System Constraints

3.1.10.1 Inference speed - AI recommendations and computer vision image scanning must return result within 3.0 seconds over standard 4g/LTE connections

3.1.10.2 Model Accuracy - Image Analysis model for health screening must maintain a minimum 85% precision Rate before Triggering Critical Health Alerts

3.1.10.3 Data Privacy - user-generated breeding predictions and farm metrics shall remain strictly isolated to the farm account and external model retaining without consent

3.1.10.4 Graceful Degradation - if AI Service or API connection become unavailable, core CRUD function (manual logging, pen assignment, and roaster view) must remaining fully operational

3.2 Usability

3.2.1 Mobile-First Responsive Design

he user interface must seamlessly adapt to mobile smartphone screens (iOS and Android browsers) with touch-friendly buttons, as farm workers primarily use phones in the field

3.2.2 Intuitive Error Handling

If a user attempts to save incomplete data (e.g., missing hatch date), the system must highlight the missing field in red and provide a clear, non-technical error message.

3.2.3 Three-Click Navigation Rule

Core actions, such as adding a new bird or logging a health treatment, must be accessible within three clicks from the main dashboard.

3.3 Reliability & Availability

3.3.1 System Uptime Guarantee

The cloud-hosted application must maintain 99.5% uptime, ensuring users can retrieve historical lineage or medical data on demand during farm hours or events.

3.3.2 Automated Backup Schedule

Database snapshots must be captured every 24 hours. In the event of a critical failure, system data must be recoverable to the most recent daily backup.

3.4 Performance

3.4.1. Respons time

Dashboard load times and search query returns must complete in under 2.0 seconds on a standard 4G/LTE mobile connection.

3.4.2. Concurrency

The database architecture must handle multiple users updating records simultaneously without data locking or corruption.

3.4.3 Calculation Speed

Age updates and win-rate percentage recalculations must execute instantly upon page refresh.

3.5 Security

3.5.1 Role-Based Access Control

Data is strictly compartmentalized. A logged-in user can only view, edit, and access records associated with their registered farm ID.

3.5.2 Data Encryption

All passwords must be hashed using bcrypt (or industry standard equivalent) prior to database storage. Data transmitted between the user's browser and the server must use HTTPS (SSL/TLS encryption).

3.5.3 *Session Timeout*

User sessions shall automatically expire after 60 minutes of inactivity to prevent unauthorized access on shared devices.

3.6 Design Constraints

3.6.1 Platform Compatibility

The system must function optimally on the latest two versions of Google Chrome, Apple Safari, and Microsoft Edge.

3.6.2 Cloud Databased Requirement

The system must utilize a robust relational database management system (RDBMS) such as PostgreSQL or MySQL to properly manage complex hierarchical relationships (lineage/pedigree).

3.7 Interfaces

Information about the interfaces

3.7.1 User Interfaces

The UI will consist of a primary navigational sidebar (or bottom nav-bar on mobile), a data-rich central dashboard displaying aggregate stats, and modal pop-ups for rapid data entry (e.g., 'Quick Add Treatment').

3.7.2 Databased Interface

The application backend will interface with the SQL database using an Object-Relational Mapper (ORM) to handle structured queries and ensure data integrity during insertions and deletions.

3.7.3 Hardware Interfaces

The web application shall request permission to interface directly with the user's mobile device camera to facilitate instant photo capture during the 'Add New Chicken' registration process.